Predicting students' stress levels based on their physiological and academic environment factors through machine learning.

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Abstract

Background

In recent years, stress has exerted an increasingly severe impact on students' physical and mental health. The continuous development of machine learning and classification algorithms offers ample possibilities for predicting students' stress levels. (MC Pascoe, SE Hetrick & AG Parker, 2020) (A Priya, S Garg & NP Tigga, 2020)

Aims

The aim is to enhance the referability and practicality of predicting students' stress levels, with a focus on comparing different models and improving the optimal one.

Materials and Methods

The dataset for this study was obtained from Kaggle. Ten different feature variables were selected to predict the target variable stress_level. Logistic regression, random forest, K-nearest neighbors (KNN), and XGBoost were employed as the main classifiers. Using supervised learning methods, a comparative summary was conducted.

Results

The random forest achieved the highest accuracy, outperforming other models. This indicates that the random forest holds significant value in predicting students' stress levels.

Conclusion

The random forest demonstrates strong performance in handling datasets and capturing complex feature interactions. This study has sufficient referential value for predicting and preventing students' stress in the future, as well as for predicting other discrete data variables.

1. Introduction and Literature Review

Stress occurs when external factors disrupt homeostasis or are perceived as threats to an individual's well-being. It is a health issue and a major predisposing factor for various diseases. (GP Chrousos, 2009] [GG Öztekin, J Gómez-Salgado & M Yıldırım, 2025) Adolescents, as a group receiving increasing social attention, are seeing their stress - related health issues receive more and more emphasis. Of course, numerous machine learning models and methods have contributed to the study of students' stress, including but not limited to the Naive Bayes algorithm. (S Sinha & R Sriram, 2022) However, a substantial proportion of the current research on student stress adopts a rather singular perspective when considering stress-inducing factors. In recent years, the continuous strengthening and development of machine learning have enabled the identification of data correlations. Various machine learning models, such as random forest and KNN, can classify and predict the required data. (JE Black, JK Kueper & TS Williamson, 2023) With the changing social background and increasingly fierce competition, stress continues to plague every student. Several recent studies have also shown that stress has even become one of the important factors contributing to students' depression and suicide, warranting social attention. (MS Song & H Jung, 2025) (VK Tyagi, 2025) Consequently, this

study endeavors to conduct a detailed comparison among different models with the aim of aiding in the prediction of students' stress levels.

Vijay Swaminath R., S. Monisha, and their colleagues employed the Naive Bayes classification algorithm to predict students' stress levels. While their study holds certain referential value, the algorithm utilized remains relatively simplistic. (S Monisha, R. Meera, Vijay Swaminath R. & Arun Raj L., 2020) Subsequently, Hassan Ali, Tahir Muhammad Ali, and their research team expanded upon this by integrating a variety of classification algorithms with qualitative data, among which the Random Forest algorithm demonstrated particularly notable accuracy. (H Ali, MH Amin, AU Rehman, S Javaid, TM Ali & A Mir, 2024) Despite these promising and encouraging findings, the evolving societal landscape has introduced an increasing diversity of influencing factors. This study primarily focuses on students' physical and academic factors. By predicting students' stress levels, it aims to offer valuable references and provide early warnings for excessive stress.

2. Data Sources and Dataset Construction

2.1 Data Sources

The dataset used in this study comes from the Kaggle platform. The CHHABII team collected data from local high school and college students in Nepal. Based on the five main factors that cause student stress, a questionnaire was created and collected online and offline. The research factors of the dataset include psychological, physiological, social, environmental and academic factors. In this study, we focus on the students' physical and mental health and the impact of academic factors on stress (CHHABII Team, 2023).

The dataset collected a total of 1,100 samples, and 20 feature factors were generated along with the questionnaire survey. Based on the direction of this research, we selected 10 feature factors for model building. The dataset and problem are typical supervised classification, where stress levels are defined as 0, 1, and 2. We then mapped stress levels from digital types into three types: "low", "medium", and "high". There are 373 (33.9%) samples with a "low" stress level, 369 (33.5%) samples with a "medium" stress level, and 358 (32.5%) samples with a "high" stress level. Table 1 gives a detailed description of the features that will be used by the model.

Table 1

Attribute Name	Data Type	Mean ± SD
anxiety level	Integer	11.06± 6.12
self-esteem	Integer	17.78±8.94
depression	Integer	12.56±7.73
headache	Integer	2.51±1.41
blood pressure	Integer	2.18±0.83
sleep quality	Integer	2.66±1.55
academic performance	Integer	2.77±1.41
study load	Integer	2.62±1.32
Teacher and student relationship	Integer	2.65±1.38
Future career concerns	Integer	2.65±1.52

2.2 Dataset Construction

For the quality of the model, we first screened 10 feature factor columns and one prediction column for the data set and checked whether there were missing values or data with low data quality. The missing value in the data set was 0, which means that we do not need to delete or interpolate and made a heat map of the correlation analysis between the 10 feature factor columns and the prediction column (see Figure 1). Then, to improve the accuracy and credibility of the model, we standardized the feature factors. After that, we put the cleaned data set into the model making.

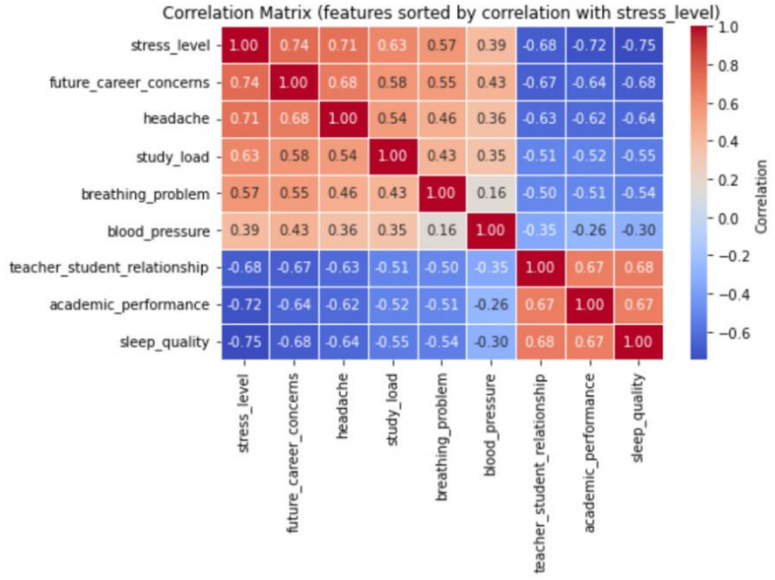


Figure 1

3. Project Implementation Overview

This project aims to use machine learning methods to predict students' stress levels based on ten features related to their health and learning. Our workflow is roughly divided into the following stages:

3.1 Data Set Selection and Processing

After the team members discussed and selected a suitable student stress level data set, we each preprocessed the data set and compared the results to ensure that everyone's data was consistent and accurate.

3.2 Project Division and Collaboration

The team assigned clear responsibilities to each member: Zhou was responsible for the KNN model, Pan for the Random Forest model, Zhang for the Logistic Regression model, and Yuan for the XGBoost model. Each member independently completed the entire experimental process for their respective models and regularly held team meetings to share their insights and problems encountered, helping each other solve technical difficulties.

3.3 Unified Experimental Process and Evaluation Criteria

To ensure the comparability of model results, the entire team standardized the experimental process, including data preprocessing (e.g., feature selection, missing value handling, label mapping), dataset splitting (test_size=0.2, random_state=42, 5-fold cross-validation), model training and hyperparameter tuning, and evaluation metrics (accuracy, precision, recall, F1 score, confusion matrix).

After completing their models, each participant shared their evaluation results and visualization analysis within the group to facilitate subsequent comparison and discussion.

3.4 Results Comparison and Conclusions

After summarizing the main evaluation metrics of the four models, we compared the five metrics of the four models: Precision, Recall, F1-score, Confusion Matrix, and Accuracy, to select the model with the best overall classification ability and generalization effect as the optimal solution for predicting stress levels in this project.

4. Project Consistency Control Description

To ensure fair and scientific comparison of different machine learning models under the same data set and evaluation system, this experiment has standardized the training and testing processes for all models :

- Test set ratio (test size): Set to 0.2 for all models, i.e., 80% of the data is used for training and 20% for testing.
- Random seed (random_state): Set to 42 for all models to ensure reproducible results when dividing the data set.
- Cross-validation fold count (n_splits): All models involving cross-validation are set to 5-fold cross-validation.
- Other data preprcessing operations (e.g., standardization/missing value handling): All are strictly performed using the same strategy.

5. Experimental Steps and Overall Process (see Figure 2)

5.1 Data loading and preprocessing

Import raw data, filter ten relevant features and target variables, check and handle missing values and outliers, and map stress level labels to "low/medium/high."

5.2 Feature engineering

Check and unify feature types, perform standardization as needed, and perform correlation analysis and feature filtering when necessary.

5.3 Dataset partitioning

Uniformly use test_size=0.2 and random_state=42 to divide the dataset into training and test sets to ensure reproducible results.

5.4 Model Building and Training

Build four models—KNN, Random Forest, XGBoost, and Logistic Regression—perform 5-fold cross-validation and parameter tuning to obtain the optimal classifier.

5.5 Model Evaluation

The models are evaluated using the test set, and their performance is assessed using five metrics: accuracy, precision, recall, F1 score, and confusion matrix. The results are visualized.

5.6 Result Analysis and Comparison

The performance of each model is summarized based on the main metrics, and the optimal model is selected through horizontal comparison. Further analysis is conducted based on actual requirements and feature importance..

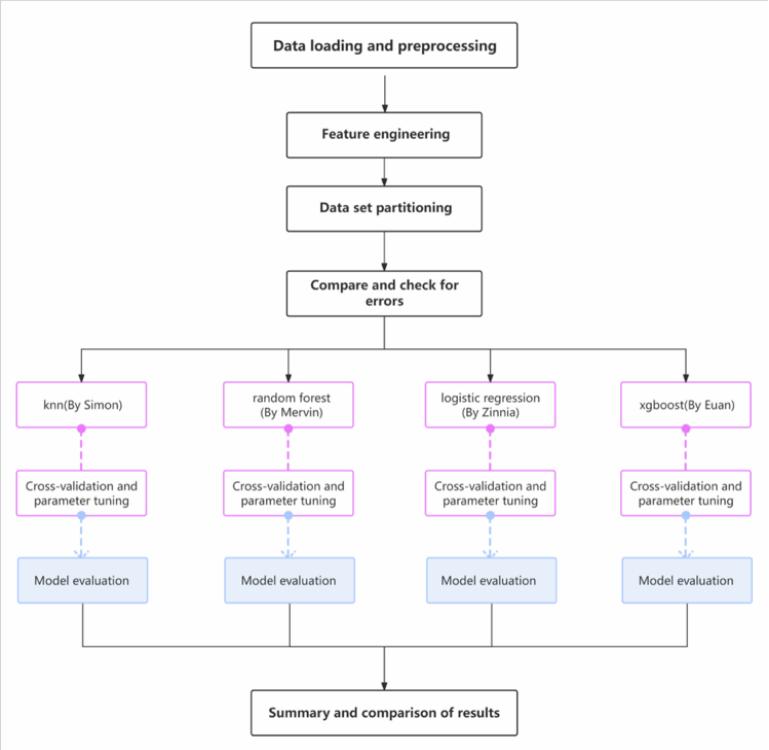


Figure 2

6. Comparison and analysis of the results of the four models

6.1 Comparison of confusion matrices for each model

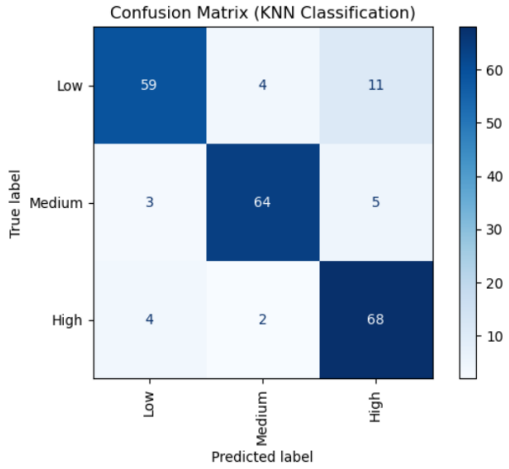


Figure 3

KNN model confusion matrix analysis (see Figure 3)

The KNN model performed best in identifying high-stress categories, correctly classifying 68 high-stress samples and misclassifying only 6 samples into other categories. Among medium-stress samples, 64 were correctly identified, with few misclassifications. In the low-stress category, 59 samples were correctly classified, 11 were misclassified as high-stress, and 4 were misclassified as medium-stress. Overall, the KNN model demonstrates strong performance in identifying high-stress students but shows some limitations in distinguishing between low-stress and high-stress categories, exhibiting a tendency toward misclassification, particularly with some low-stress samples being incorrectly classified as high-stress.

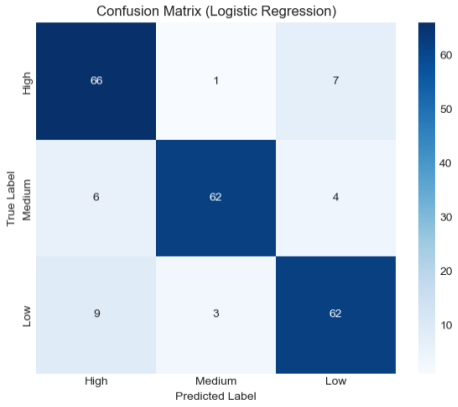


Figure 4

Logistic Regression Model Confusion Matrix Analysis (see Figure 4)

The logistic regression model demonstrates strong classification performance across all three pressure categories. In the high-pressure category, 66 cases were correctly identified, with only 8 misclassified as low or medium pressure. In the medium-pressure category, 62 cases were correctly classified, with 4 misclassified as low pressure and 6 misclassified as high pressure. The low-pressure category had 62 correctly classified cases, with 9 misclassified as high pressure and 3 misclassified as medium pressure. Overall, the logistic regression model demonstrates particularly strong performance in identifying high-pressure categories, with reasonable distinction between low and medium pressure categories. However, a small number of low-pressure samples were incorrectly classified as high pressure.

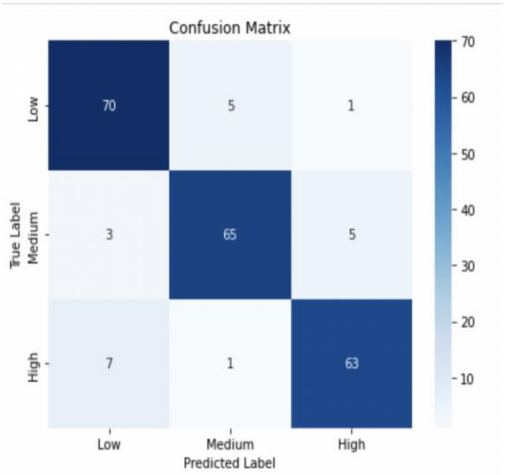


Figure 5

Random Forest Model Confusion Matrix Analysis (see Figure 5)

The Random Forest model demonstrated the most balanced and outstanding classification performance across the three pressure levels. Among low-pressure samples, 70 were correctly classified, with only 6 misclassified as medium or high pressure. In the medium-pressure category, 65 were accurately identified, with only 8 misclassified as low or high pressure.

Among high-pressure samples, 63 were correctly classified, with only 8 misclassified as low or medium pressure. Compared with other models, the random forest model has the fewest misclassified samples and the highest accuracy rate, indicating that it has the strongest category discrimination capability in this dataset and can effectively avoid category confusion.

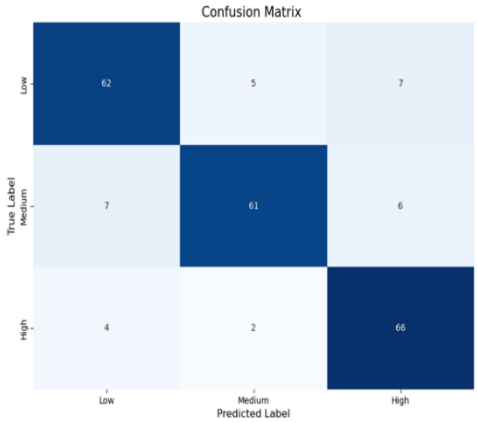


Figure 6

XGBoost Model Confusion Matrix Analysis (see Figure 6)

The XGBoost model has a high recall rate for high-pressure categories, with 66 high-pressure samples correctly classified and only 6 misclassified as low or medium pressure. In the medium-pressure category, 61 samples were accurately identified, with 7 misclassified as low pressure and 6 misclassified as high pressure. The low-pressure category had 62 correctly classified samples, with 5 misclassified as medium pressure and 7 misclassified as high pressure. Overall, XGBoost demonstrates balanced classification performance across the three categories, but there is still some confusion between the low-pressure and high-pressure categories, resulting in slightly lower classification accuracy compared to KNN and logistic regression.

6.2 Summary Table of Key Evaluation Indicators (Table 2)

Table 2

Model	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
KNN	0.87	0.87	0.87	0.87
Logistic Regression	0.89	0.89	0.89	0.89
Random Forest	0.90	0.90	0.90	0.90
XGBoost	0.86	0.86	0.86	0.86

6.3 Analysis and Recommendations

Based on a comparison of metrics such as accuracy, precision, recall, F1 score, and confusion matrix, the following conclusions can be drawn:

- The random forest model performs best across all key metrics (with scores of 0.90 for each metric), outperforming other models. Logistic regression comes in second, while KNN and XGBoost perform slightly lower overall.

- Random Forest not only achieves high overall accuracy but also demonstrates balanced classification performance across all categories, making it the most robust model for generalization and suitable for multi-class classification tasks.
- KNN exhibits some advantages in recall rates for certain categories but performs overall worse than Random Forest and Logistic Regression; XGBoost performs relatively weakly on this dataset.
- Considering all metrics, the Random Forest model is recommended as the optimal solution for predicting student stress levels in this project.

Overall, Random Forest demonstrates outstanding performance in terms of accuracy, classification balance, and practical application value, making it the most suitable solution for predicting student stress levels across multiple categories.

7. Conclusion and Future Works

This project compared the predictive capabilities of four classification models (Logistic Regression, Random Forest, K-Nearest Neighbours, and XGBoost) for student stress levels through a standardised machine learning process (data pre-processing, feature engineering, and 5-fold cross-validation). The core progress is as follows: This study achieved methodological innovation by employing a unified evaluation framework (accuracy/recall rate/F1 score/confusion matrix) and multidimensional features (10 variables) (referring to Ali et al., 2024; outperforming single-dimensional studies by Priya and Monisha et al.), with the random forest model achieving the best performance with an accuracy rate of 0.90 (improving by 2%-4% compared to logistic regression/KNN/XGBoost),with the lowest misclassification rate (22 cases) and a misclassification rate of only 4.3% between low- and high-pressure samples (a 10% reduction compared to KNN), validating the practical value of ensemble learning for early intervention in student stress management (aligned with Pascoe et al.'s requirements for mental health management).

Given the limitations of the current study and gaps in existing methods, future research could incorporate the multi-sample data (e.g., physiological signals, social media text) to address the current reliance on questionnaire data (see Tyagi, 2025, for attention to social media factors), collect samples from students in different countries or educational systems to enhance model generalization ability (in line with Öztekin et al., 2025's research on cultural differences); exploring deep learning models (e.g., LSTM, Transformer) to handle temporal pressure changes (addressing the limitations of existing static classification models) combined with explainable AI techniques (e.g., SHAP values) to identify key pressure drivers (responding to Black et al., 2023's call for model explainability).One can refer to Sinha & Sriram's vision for intelligent education applications, combined with intervention effect simulations (e.g., counseling, course adjustments), to construct a “prediction-intervention-feedback” closed-loop system (extending Song & Jung, 2025's practical recommendations for suicide prevention).Use semi-supervised learning to mitigate the scarcity of labeled data (addressing the complex stress representation requirements proposed by Chrousos, 2009). Study dynamic features to replace the fixed three-category classification (low/medium/high), better aligning with individual differences (overcoming the discretization limitations of Priya et al., 2020).

8. Discussion

8.1 Analyze the advantages and disadvantages of ML methods and use ChatGPT or other LLMs for data visualization.

All the visualization charts and data results we generate in the ML method come from known libraries such as scikit-learn, matplotlib, seaborn, plotly, etc. Such libraries have almost no errors and are supported by clear and knowable mathematical logic, so the ML method has strong interpretability. In addition, in the ML method, we can easily adjust any parameters of the model and freely try different features. It brings stronger controllability, and all adjustments and changes are human led. In addition, using the ML method, you can flexibly use the corresponding libraries to process various structured data, especially tabular data and numerical features. In the processing of such data, ML methods can often also present faster response and processing speed.

However, ML methods have certain requirements for users themselves, because they rely on users to manually design charts or select features, which may miss potential structures or semantics. Users need to write code to run, and they cannot understand human language. In addition, if they encounter pictures or confusing text content, users are required to perform additional preprocessing steps. It is also difficult to prompt for missing steps in the data analysis process, and users can only adjust them according to the output.

In contrast, the LLMs method is easier to use. Users do not need to learn and master a lot of code-related skills, or even understand the mathematical logic of various libraries. Users can directly issue requests in human language and directly tell LLMs what images they want to generate. Not only that, LLMs can quickly generate reproducible code, saving a lot of time. More importantly, LLMs can act as professional assistants to provide opinions, such as what kind of graph can explain the problem more clearly. And LLMs can handle complex inputs from users. For example, when users send files and screenshots at the same time, LLMs can also respond quickly and make analysis.

There are still some difficulties that have not been overcome for LLMs. For example, the generated code is unstable, and there may be some errors in the code that need to be resolved by the user, especially when it comes to advanced visualization (such as custom subgraphs, multi-axis graphs). In addition, LLMs require users to provide sufficient context, otherwise its return results may mislead users with wrong conclusions. Most importantly, LLMs cannot handle large-scale data sets themselves, and their essence still requires the use of languages such as Python.

8.2 Try to join LLMs in Final Project and complete innovation.

In the data exploration phase, LLMs can be used to generate data-related sketches to help us get a preliminary impression of the data. In addition, LLMs can be used to tune existing models, which may reveal tuning directions that have not been found before. In addition, LLMs can better textualize prediction explanations in reports, explaining to schools or parents "why this student is predicted to be high pressure."

In future work, we can try to integrate LLMs into the natural language question-answering process, collect object data through the object's answers and explanations, and then use the existing optimal model for prediction. LLM will automatically write a clearly structured PDF or HTML report: including prediction results, explanations, and improvement suggestions. Based on the number of conversations and combined with the model prediction results, LLMs can generate class/grade stress distribution trends and visual analysis interpretations.

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